

DURGAM KOUSHIK

Aspiring VLSI Physical Design Engineer
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SUMMARY

Entry-level ECE graduate with foundational knowledge of digital design, CMOS fundamentals, and the RTL-to-GDSII flow. Hands-on exposure to floorplanning, placement, CTS, routing and STA using ICC2 on a 45nm design. Seeking an entry-level Physical Design role to apply skills in timing closure, and PPA optimization.

EDUCATION

B.Tech / B.E. in Electronics and Communication Engineering

2022 – 2026

Nalla Narasimha Reddy Education Society's Group of Institutions , Hyderabad

CGPA: 8.01 / 10

Senior Secondary (Class XII), Narayana Junior college, Hyderabad | TSBIE

2020

Secondary (Class X), Narayana High School, Hyderabad | TS BSE

2018

TECHNICAL SKILLS

VLSI / Physical Design: Floorplanning, Placement, Clock Tree Synthesis (CTS), Routing, Static Timing Analysis (STA), Timing Closure, Physical Verification (DRC, LVS), PVT Corners, Setup and Hold Timing, Power Performance and Area (PPA), SDC / Timing Constraints

HDL: Verilog,

Scripting: TCL

EDA Tools: Synopsys ICC2

Operating Systems: Linux

Core Electronics: Digital Electronics, CMOS Fundamentals, Semiconductor Device Fundamentals

VLSI PROJECTS

RAVEN WRAPPER

Tools/Technology: IC Compiler II (ICC2) *Technology Node:* 45nm]. Performed backend physical design of a RAVEN WRAPPER block from synthesized netlist to routed layout as part of the RTL-to-GDSII flow.

- Performed floorplanning, power planning, placement using ICC2 to establish core area, macro placement, and power grid structure.
- Carried out clock tree synthesis, routing and analyzed post-CTS/post-route timing reports for setup and hold violations.
- Ran Static Timing Analysis (STA) across and applied SDC constraints to guide timing closure.
- Performed physical verification including DRC and LVS checks to ensure layout correctness; measured result: WNS/TNS, area utilization, or leakage power .

CNTFET-Based Booth Multiplier for Low-Power Applications

- Designed a Carbon Nanotube FET (CNTFET) based Booth multiplier for low-power VLSI applications.
- Got a good reduction in power consumption, propagation delay, and Power-Delay Product (PDP) compared to the CMOS version.
- Simulated the design in Cadence Virtuoso and studied transistor-level switching behaviour.

Low-Power 4-Bit Array Multiplier in CMOS Technology

- Designed a 4-bit array multiplier in Cadence Virtuoso using optimized adder architectures for better power and area.
- Used reduced-transistor full adder cells; verified the design through schematic simulation and layout.
- Published this work in a Scopus-indexed journal (details under Publications).

INTERNSHIP / TRAINING

VLSI, Physical Design – Takshila Institute of VLSI technologies

December, 2025 – August, 2026

- Completed structured training in Physical Design covering floorplanning, placement, CTS, routing, STA .
- Gained hands-on lab exposure to ICC2 on a 45nm design as part of the RTL-to-GDSII flow.

PUBLICATIONS

- Durgam Koushik, Dubbaka Srija Goud, and Dayala Mahesh — "Reduced Transistor-Based Full Adder Architecture for a Power Efficient 4-Bit Array Multiplier", Kronika Journal (Scopus Indexed), ISSN: 0023-4923, 2026.

CERTIFICATIONS

- NPTEL – Introduction to IoT, IIT Kharagpur (Elite + Silver Medal)
- Microchip Technology Certification, AICTE – EduSkills

ACHIEVEMENTS

- 2nd Place – AR & VR Internal Hackathon (out of 22 teams), NNRG
- Best Performer Award – Student Development Program on IoT and Embedded Systems, Team NIVA

EXTRACURRICULAR ACTIVITIES

- Event Organizer & Coordinator for college technical fests; NSS volunteer; volunteered at IIC, IIIC and IEEE Chapter events

DECLARATION

I hereby declare that the above information is true and accurate to the best of my knowledge. — Koushik | Place: Hyderabad | Date: